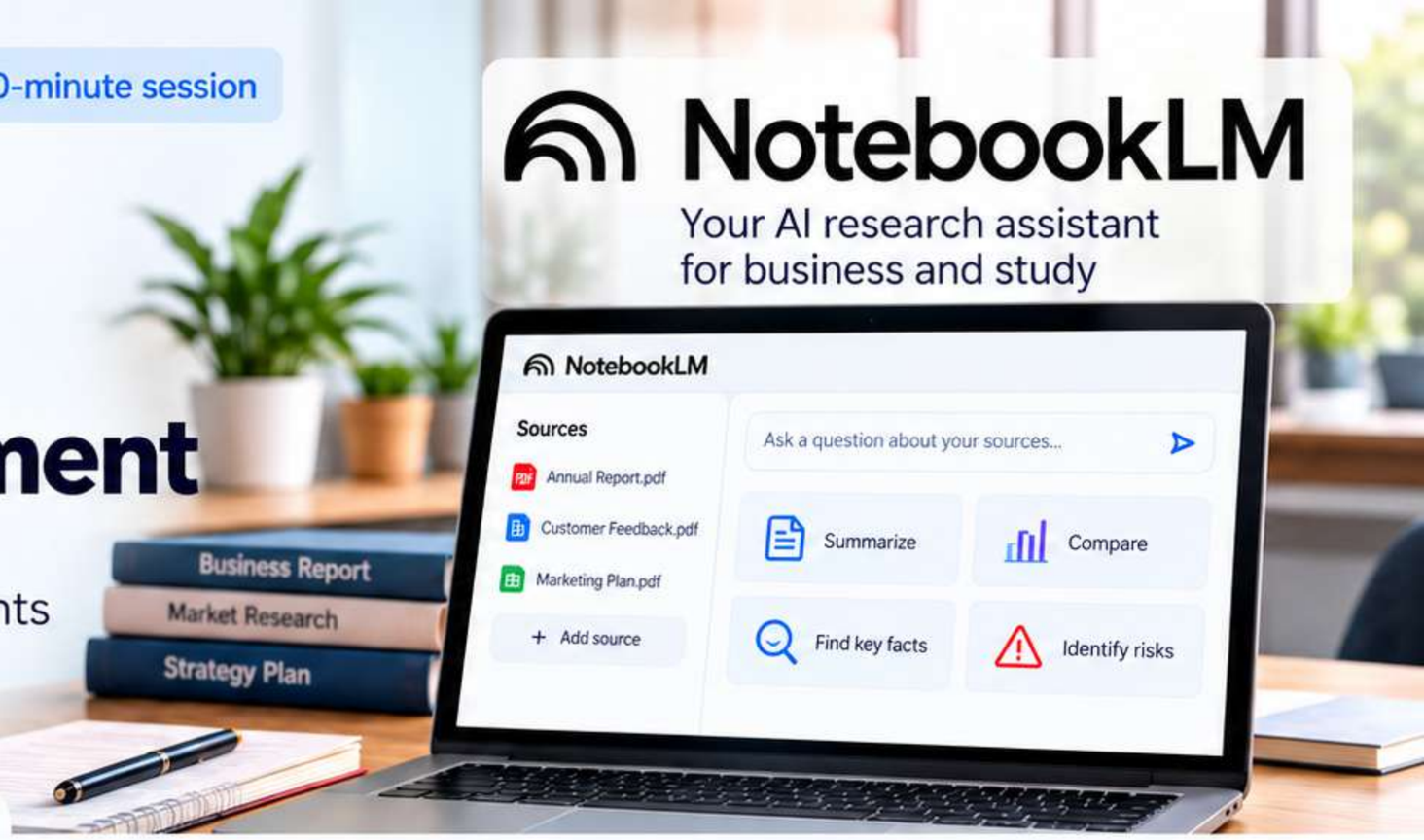
AI & Generative AI for Business • BBA 2nd Year

50-minute session

Session 34
NotebookLM & Knowledge Management

Build a source-grounded AI knowledge base that answers questions using specific documents and helps businesses turn information into useful briefings.



Learning objectives

By the end of this session, students will be able to:

1. Explain source-grounded AI and knowledge management.
2. Upload and organize business documents in NotebookLM.
3. Summarize documents and ask questions about their contents.
4. Create a research workflow using multiple sources.
5. Verify AI answers against citations and reduce hallucinations.

How NotebookLM works

1 Provide sources

PDF

Upload reports, PDFs, notes, or other supported business documents.

2 Ask questions

Summarize, compare, extract facts, and identify key findings.

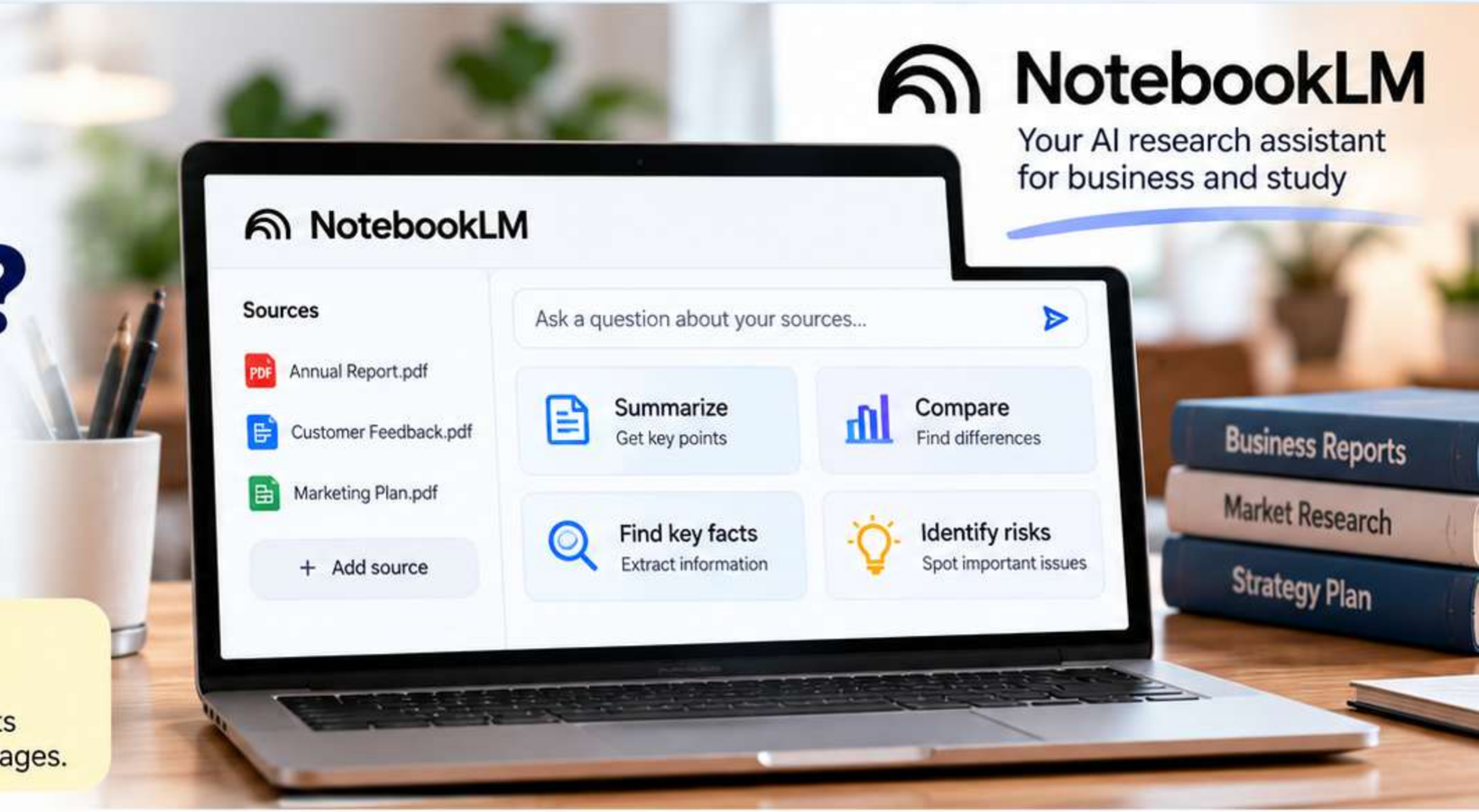
3 Create an output

Prepare a briefing, research notes, or a management summary with source references.

2. What is NotebookLM?

What is NotebookLM?

NotebookLM is an AI-powered research assistant that works with the documents you provide. It helps you summarize, analyze, and answer questions using specific source material.



Key idea

Instead of relying only on generad AI knowledge, NotebookLM focuses on your selected documents and can connect answers to relevant source passages.

How NotebookLM works

1 Provide sources
Upload reports, PDFs, notes, or other supported business documents.

PDF

2 Ask questions
Summarize, compare, extract facts, and identify key findings from your sources.

What are the main findings in these reports?

3 Create an output
Prepare a briefing, research notes, or a management summary.

- Summary report
- Key insights
- Recommendations

3. Source-Grounded AI vs. General AI

Source-Grounded AI vs. General AI

Both are useful, but they work differently. NotebookLM focuses on the documents you provide and connects answers to specific source passages.

General-purpose AI

Uses broad knowledge and your prompt to generate answers.



Can draw on broad knowledge
Uses trained knowledge and your prompt.

May answer beyond your documents
Can include information not in your sources.

Useful for brainstorming
Good for general explanations and ideas.

Requires fact-checking
Answers may not be connected to specific documents.

VS

Source-grounded AI (NotebookLM)

Focuses on the documents you provide and connects answers to relevant passages.



Focuses on selected source material
Uses only the documents you upload.

Connects answers to source passages
Shows citations or relevant sections from your documents.

Useful for document research
Helps summarize, compare, and extract key information.

Still requires verification
Reduces hallucinations, but you should check citations and source quality.

Important note
Source grounding can reduce unsupported answers, but it does not eliminate hallucinations. Missing information, ambiguous documents, or incorrect source data can still lead to misleading results.

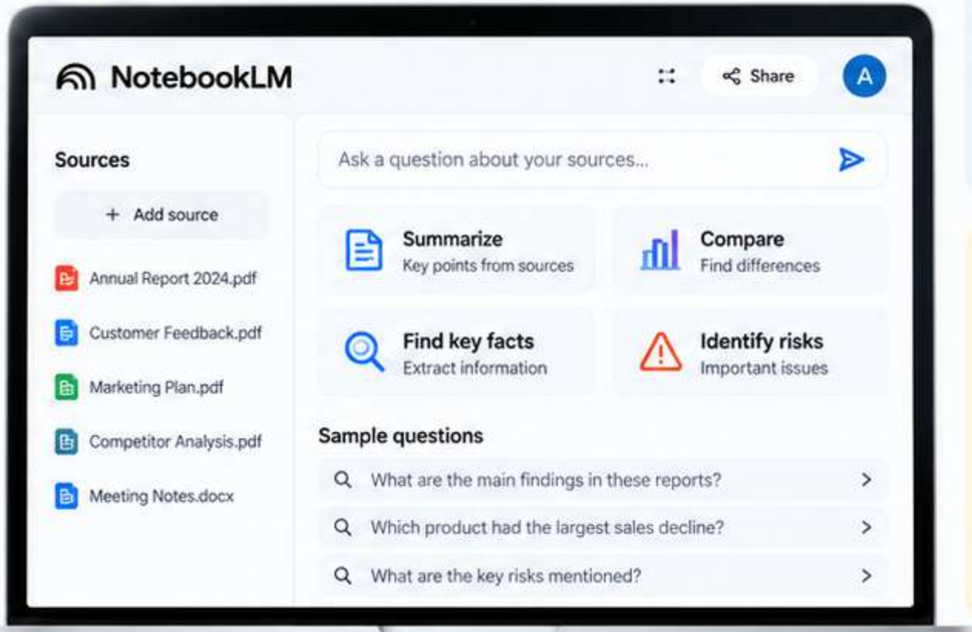
Key takeaway
NotebookLM is particularly useful when answers must stay connected to specific documents, but reliable insights depend on accurate sources, careful verification, and human judgment.

4. Uploading and Organizing Business Documents

How to build a knowledge base in NotebookLM

Create a notebook, upload your documents, and organize them so you can easily ask questions and get useful, source-grounded answers.

1. **Open NotebookLM**
   Visit notebooklm.google.com and sign in with an eligible Google account.
2. **Create a new notebook**
   Create a separate notebook for one project, course, department, or business topic.
3. **Add business documents**
   Upload supported PDFs, reports, meeting notes, or other available source types.
4. **Review the sources**
   Check that files are readable, complete, relevant, and the correct versions.
5. **Ask source-based questions**
   Ask for summaries, key figures, comparisons, risks, and evidence from the sources.
6. **Create a briefing**
   Turn verified findings into a concise report with source references.



Supported file types

- **PDF** — Reports, Documents
- **Google Docs** — Notes, Documents
- **Google Sheets** — Data, Tables
- **Word** — Reports, Documents
- **PowerPoint** — Presentations, Slides
- **Text files** — Notes, Plain text

Good practices

- Use clear and relevant documents
- Keep file names organized
- Remove duplicate or outdated files
- Maintain one notebook per project/topic
- Check that documents are complete and readable

Example

Annual Report.pdf + Customer Feedback.pdf + Marketing Plan.pdf

→ "Summarize the main findings, compare this year with last year, and identify key risks with source citations."

→ **AI output**

- Summary
- Key insights
- Citations
- Recommendations

Key takeaway: A well-organized notebook with relevant and reliable documents helps you get more accurate and useful answers from NotebookLM.

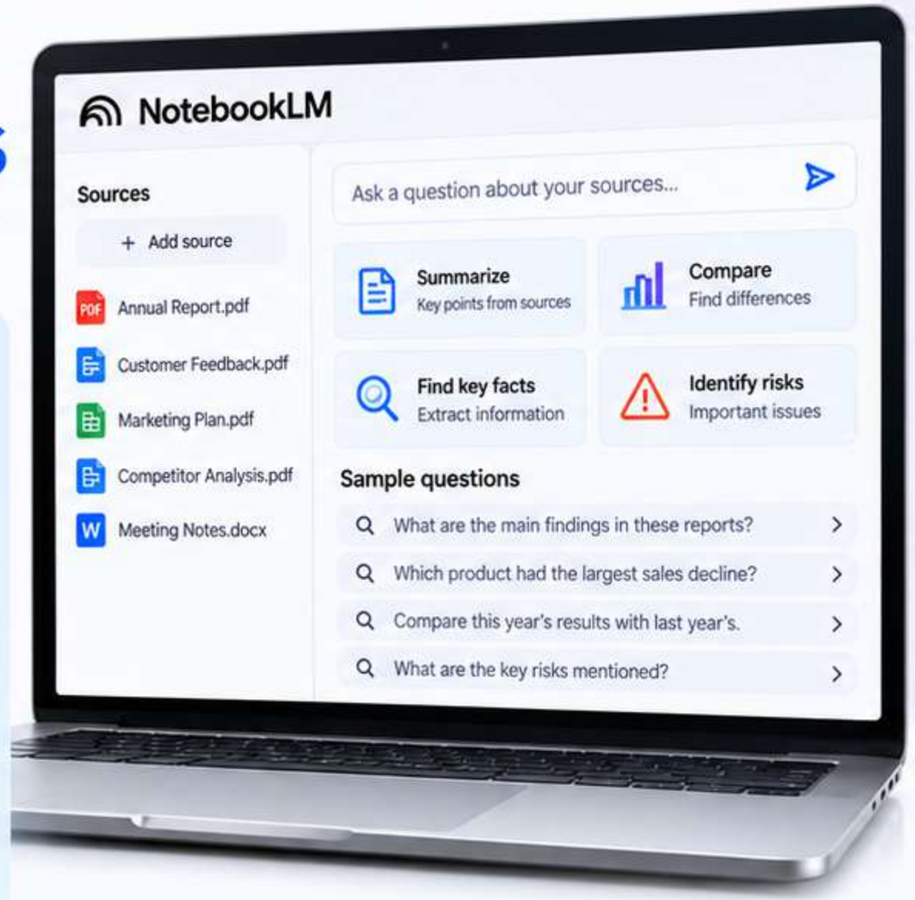
5. Asking Better Questions and Summarizing Documents

Get useful answers from your documents

Ask clear, specific, and focused questions to get better summaries, analysis, and insights from NotebookLM.

Types of questions you can ask

1. **Summarization**
   "Summarize the report in five key points."
2. **Fact extraction**
   "What marketing budget is mentioned in the documents?"
3. **Comparison**
   "Compare this year's targets with last year's results."
4. **Risk identification**
   "What risks are mentioned in the reports?"
5. **Evidence gathering**
   "Which source supports this conclusion?"
6. **Gap analysis**
   "What information is missing to make this decision?"



Example prompts for business use

Business summary
"Summarize the main findings from these three reports in five bullet points."

Financial analysis
"What is the total revenue, total expense, and profit for this year?"

Comparison
"Compare this year's performance with last year's and highlight key differences."

Risk analysis
"Identify the top three business risks mentioned in these documents."

Action plan
"Based on these documents, suggest three practical next steps for management."

Example: Summarizing business documents

PDF Annual Report.pdf | Customer Feedback.pdf | Marketing Plan.pdf

→

NotebookLM

Summary (example output)

1. Overall performance improved by 12%.
2. Customer satisfaction declined due to delivery delays.
3. Marketing budget increased in Q2.
4. Key risks include rising costs and high competition.
5. Recommended focus areas are product quality, delivery time, and customer support.

Source 1 Source 2 Source 3

Good question tips

- Be clear and specific
- Mention the documents you want to use
- Ask for citations or source references
- Request tables or structured answers
- Ask for a summary in a specific format

Key takeaway

Better questions lead to more relevant and reliable answers, but always verify the results with the original documents.

6. Research Workflow and Business Briefings

A Practical Research Workflow with NotebookLM

Turn multiple business documents into clear insights and professional briefings using a simple step-by-step workflow.

1 Collect sources

Gather relevant documents for your topic.

- Reports and annual statements
- Research papers
- Meeting notes
- Policies and guidelines

2 Organize and inspect

Check the relevance, quality, and completeness of your sources.



- Are the files relevant?
- Are they up to date?
- Are they complete and readable?
- Are the correct versions used?

3 Ask targeted questions

Use clear and specific questions to extract useful information.

- What are the main findings?
- Compare this year with last year.
- What risks are mentioned?
- Which source supports this?

4 Verify and analyze

Check citations and review the original sources.



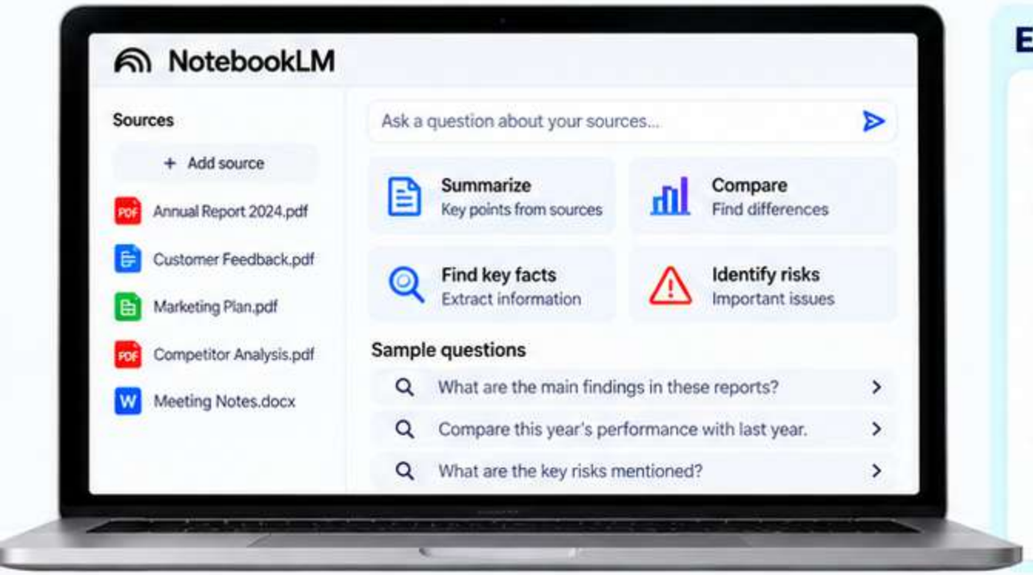
- Read the cited passage
- Verify the figures and context
- Check for missing or conflicting information
- Distinguish facts from inferences

5 Create a briefing

Turn verified findings into a concise report.



- Key findings
- Evidence from sources
- Business implications
- Recommended actions



Example output: Business briefing

Management Briefing

Based on: Annual Report 2024, Customer Feedback, Marketing Plan

Key findings

- Overall revenue increased by 12% compared to last year.
- Customer satisfaction declined due to delivery delays.
- Marketing budget increased in Q2, leading to higher website traffic.

Top business risks

- Rising operational costs.
- Increased customer complaints.
- Potential market competition in the next quarter.

Recommended next steps

- Improve delivery process and reduce average delivery time.
- Focus marketing on high-performing product categories.
- Monitor cost increases and explore alternative suppliers.

Source references

Source 1 (p. 5)
Source 2 (p. 3)
Source 1 (p. 12)
Source 3 (p. 4)
Source 2 (p. 7)
Source 1 (p. 18)
Source 2 (p. 10)
Source 3 (p. 6)
Source 1 (p. 20)

Classroom activity — 15 minutes

Work in small groups to create a source-grounded business briefing using documents provided by the teacher.

1. Upload at least two business documents.
2. Ask five relevant questions.
3. Verify at least three answers with citations.
4. Create a one-page management briefing.

Key takeaway

A clear research workflow helps you get more accurate and useful insights from NotebookLM, but reliable results depend on good sources and careful verification.

7. Hallucinations, Citations, and Verification

Use AI responsibly with reliable sources

Source grounding can reduce unsupported answers, but AI can still make mistakes. Always verify citations and check the original documents.



What is a hallucination?

A hallucination occurs when an AI produces information that sounds plausible but is unsupported or incorrect.

Why can it happen?

- Missing information in the sources
- Ambiguous or unclear documents
- Outdated or incorrect data
- Conflicting information across documents

Unsupported / Hallucinated Answer

"Customer satisfaction declined mainly because of higher prices and increased competition from new online platforms."

- Sounds reasonable, but not supported by the uploaded documents.
- No citations or evidence provided.

Source-grounded Answer (Better)

"According to the Customer Feedback report (p. 3), 62% of customers mentioned delivery delays. The Marketing Plan (p. 12) also notes product quality issues as a key concern."

- Connected to specific source passages.
- Includes citations for verification.
- Still needs human review to confirm accuracy and context.

Good practices to improve reliability

1. **Limit the scope**
   Tell the AI to use only the supplied documents.
2. **Ask for evidence**
   Request citations or the relevant passage.
3. **Verify key facts**
   Open the cited source and confirm the figures and context.
4. **Check source quality**
   Look for outdated, conflicting, or incomplete information.
5. **Allow uncertainty**
   Ask the AI to state when the answer is not available.
6. **Protect confidential data**
   Follow institutional policies before uploading sensitive information.

Example: Verify a citation

"The marketing budget increased by 20% in 2024."

Source 1 (p. 8)

Page 8

Marketing Plan 2024

The marketing budget increased by 20% in 2024 to support digital campaigns and campus outreach.

Example of a better prompt

"Using only the documents in this notebook, summarize the main findings in five bullet points. Identify the top three business risks with citations. If information is missing or conflicting, explain the gap instead of guessing."

Key takeaway: Source-grounded AI is particularly useful when answers must stay connected to specific documents. However, trustworthy insights still depend on reliable sources, careful verification, and human judgment.

8. Practical Activity: Build a Source-Grounded Knowledge Base

From Documents to a Management Briefing

Work in small groups to use NotebookLM with real business documents and create a concise, evidence-based briefing for management.



1 Create a notebook

Make a new notebook for your project or topic.

NotebookLM

2 Upload documents

Add at least two relevant business documents.

PDF

3 Ask questions

Use clear and specific questions to get useful insights.

- Summarize the findings.
- Compare this year with last year.
- What are the key risks?

4 Verify the answers

Check the citations and read the original passages.

5 Create a briefing

Turn verified findings into a one-page management report.

Activity checklist

- [ ] Create a NotebookLM notebook.
- [ ] Upload at least two relevant business documents.
- [ ] Ask five questions about the documents.
- [ ] Verify at least three answers against their citations.
- [ ] Prepare a one-page management briefing.
- [ ] List one limitation or missing piece of information.

Example documents (subject to availability)

Annual Business Report.pdf
Revenue, expenses, performance, and targets

Customer Feedback.pdf
Complaints, satisfaction, and suggestions

Marketing Plan.pdf
Campaign budget, audience, and objectives

Suggested prompt for this activity

"Using only the documents in this notebook, summarize the main findings in five bullet points. Identify the top three business risks, support each point with source citations, and suggest three practical next steps for management. If information is missing or conflicting, explain the gaps instead of guessing."

Expected output

Key findings"
Summary from sources

Source references
Citations for each claim

Business implications
What it means for the business

Recommended actions
Practical next steps

Discussion question

What could happen if a business uploads outdated or incorrect information into its knowledge base?

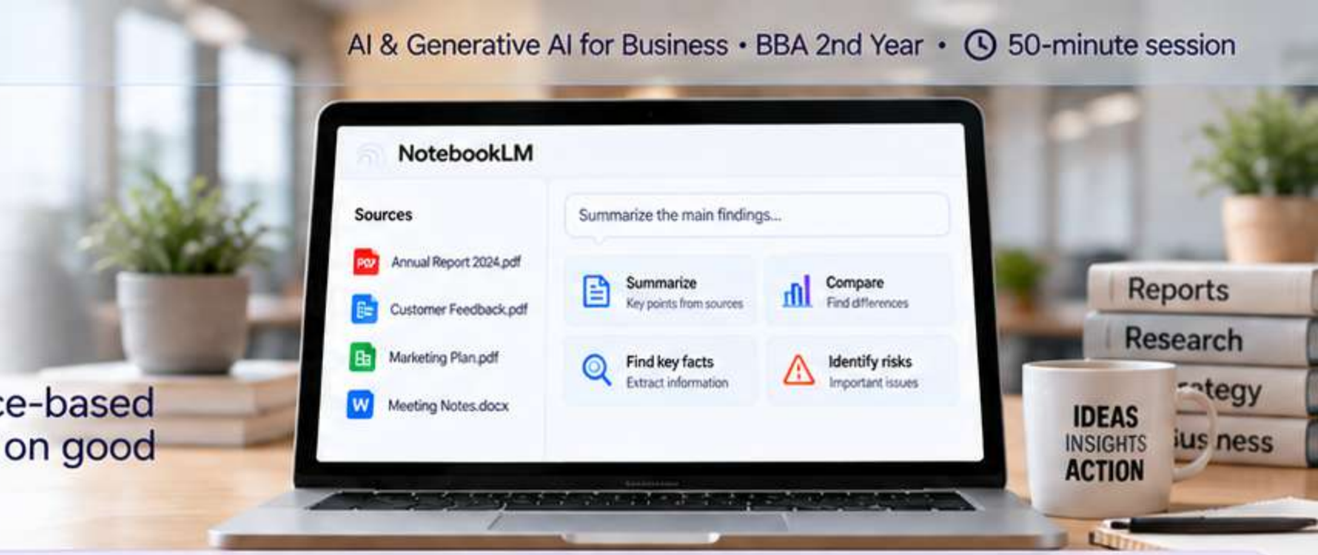
Key takeaway

A structured workflow helps you get more useful insights from NotebookLM, but reliable conclusions depend on good sources and careful verification.

9. Key Takeaways and Session Recap

Turn Information into Useful Business Insights

NotebookLM helps you research, summarize, and create evidence-based briefings using your own documents, but reliable results depend on good sources and careful verification.



Key takeaways from this session

1. **Source-grounded AI**
   Uses your uploaded documents to provide answers with relevant source references.
2. **Document management**
   Upload, organize, and review business documents for better results.
3. **Research and analysis**
   Summarize, extract facts, compare information, and identify risks using targeted questions.
4. **Verification is essential**
   Always check AI answers against the original documents. Citations help, but can still be incorrect.
5. **Practical business use**
   Create clear, evidence-based briefings for management decisions, research, and reporting.
6. **Limitations and responsibilities**
   Results depend on the quality of sources and human judgment. Follow data privacy and institutional policies.

Final key takeaway

> "Source-grounded AI is particularly useful when answers must stay connected to specific documents."

Quick knowledge check

4 questions

1. What makes source-grounded AI useful?

- A. It always gives correct answers
- **B. It connects answers to supplied sources**
- C. It never needs verification

2. What should you do when an AI answer includes a citation?

- A. Assume it is correct
- **B. Verify the claim in the cited source**
- C. Ignore the source

3. What should AI do when the documents lack the answer?

- A. Invent a likely answer
- **B. State that the information is missing**
- C. Use an unsupported statistic

4. Why is it important to use accurate and up-to-date documents?

- **A. It reduces the risk of unreliable or misleading answers**
- B. It makes AI faster
- C. It guarantees correct predictions

Next steps / Assignment

1. Choose a business topic and collect 2–3 relevant documents.
2. Build a NotebookLM knowledge base and ask at least five questions.
3. Prepare a one-page business briefing with key findings, citations, recommendations, and at least two limitations or information gaps.

